Program:

```
clear; clc;
xdel(winsid());//close all windows
A=[0 1;-0.4 -1.3];
B=[0;1];
C=[0.8 1];
D=[0];
G1=syslin('c',A,B,C,D); ssprint(G1);
G2=syslin('c',A',C',B',D);ssprint(G2);
Cc1=cont_mat(A,B);disp(Cc1,'state controllability matrix1=');
disp(det(Cc1),'det(Cc1)=');
if(det(Cc1)~=0)
printf('The given system 1 is completely controllable')
else
printf('The given system 1 is not completely controllable')
end
Ob1=obsv_mat(A,C);disp(Ob1,'observability matrix1=');
disp(det(Ob1),'det(Ob1)');
if(det(Ob1)~=0)
printf('The given system 1 is completely observable')
else
printf('The given system 1 is not completely observable')

end
Cc2=cont_mat(A',C');disp(Cc2,'state controllability matrix2=');
disp(det(Cc2),'det(Cc2)=');
if(det(Cc2)~=0)
printf('The given system 2 is completely controllable')
```

```
else
printf('The given system 2 is not completely controllable')
end
Ob2=obsv_mat(A',B');disp(Ob2,'observability matrix2=');
disp(det(Ob2),'det(Ob2)');
if(det(Ob2)~=0)
printf('The given system 2 is completely observable')
else
printf('The given system 2 is not completely observable')
end
Htf=ss2tf(G1);disp (Htf,'Reduced transfer function=');
e=spec(A); disp (e,'Eigen values=');
D=poly(e,'s'); disp(D,'actual denominator (characteristic poly)=');
//Htf=ss2tf(G1);disp (Htf,'Reduced transfer function=');
//e=spec(A); disp (e,'Eigen values=');
//D=poly(e,'s'); disp(D,'actual denominator (characteristic poly)=');
```

OUTPUT:

```
  [state-space]
  A (matrix) = [0,1;-0.4,-1.3]
  B (matrix) = [0;1]
  C (matrix) = [0.8,1]
  D (matrix) = 0
  X0 (initial state) = [0;0]
  dt (time domain) = "c"
  [state-space]
  A (matrix) = [0,-0.4;1,-1.3]
  B (matrix) = [0.8;1]
  C (matrix) = [0,1]
  D (matrix) = 0
  X0 (initial state) = [0;0]
  dt (time domain) = "c"
   0.   1.
   1.  -1.3
  "state controllability matrix1="
  -1.
  "det(Cc1)="
The given system 1 is completely controllable
   0.8   1.
  -0.4  -0.5
  "observability matrix1="
   0.
  "det(Ob1)"
The given system 1 is not completely observable
   0.8  -0.4
   1.   -0.5
  "state controllability matrix2="
   0.
  "det(Cc2)="
The given system 2 is not completely controllable
   0.   1.
   1.  -1.3
  "observability matrix2="
  -1.
  "det(Ob2)"
The given system 2 is completely observable
     1
```

```
The given system 2 is completely observable
     1
   ------
   0.5 +s
  "Reduced transfer function="
  -0.5 + 0.i
  -0.8 + 0.i
  "Eigen values="
  0.4 +1.3s +s^2
  "actual denominator (characteristic poly)="
```